Question	Answer	Marks	Partial Marks	Guidance
10(a)	$3x^2 - 18x - 48$ final answer	**2**	**B1** for two correct terms or for correct answer seen then spoiled	Accept equivalents for B2 or B1 e.g. B2 for $3(x^2 - 6x - 16)$ Condone poor notation for B2 or B1 e.g. $y = 3x^2 - 18x - 48$, $y = 3x^2 - 18x$
10(b)	*their* $\frac{dy}{dx} = 0$ soi	**M1**		After 2 marks in (a) then must use correct derivative = 0 to score M1 M1 can be implied by a correct method without errors to find x or for answers $x = -2$ and $x = 8$ FT *their* dy/dx
	$[3](x - 8)(x + 2)$ oe or $\frac{-(-18) \pm \sqrt{(-18)^2 - 4(3)(-48)}}{2 \times 3}$ oe or $3 \pm \sqrt{16 + 3^2}$	**M1**	oe $\frac{18 \pm \sqrt{900}}{6}$	FT *their* 3-term quadratic oe eg $(3x - 24)(x + 2)$ or $(3x + 6)(x - 8)$ Allow B1 for correct $(ax + b)(cx + d)$ from *their* quadratic Allow 18^2 and allow -18^2 if recovered. Allow 18 for $-(-18)$ **No recovery for short line or short root unless seen or implied in working** Fraction line must underscore the ± sign Length of root must cover – sign, ie reach 4 or complete square method correct for *their* quadratic
	(–2, 52) (8, –448)	**B2**	**B1** for one correct pair of coordinates or for two correct values of x	

Question	Answer	Marks	Partial Marks	Guidance
10(c)	(–2, 52) maximum with reason and $(8, -448)$ minimum with reason and no incorrect statement	**3**	Reasons could be eg 1. A reasonable sketch of a positive cubic 2. Correct evaluation and use of 2nd derivative $6x - 18 = -30$, $-30 < 0$, so (–2, 52) is a maximum oe. $6x - 18 = 30$, $30 > 0$, so $(8, -448)$ is a minimum oe. 3. Evaluates correctly values of y on both sides of both correct stationary points 4. Finds gradient on each side of both correct stationary points. Any incorrect statement MAX B2 **B2** for 1 correct with correct reason for that stationary point or for **both** x-values correct and reasonable sketch of a positive cubic or for correct substitution and evaluation of **both** of *their* x-values into *their* second derivative or substitution and evaluation for one x-value on both sides of **both** of *their* stationary points to find the gradients soi or **M1** for showing [2nd derivative =] $6x - 18$ or correct FT *their* 2nd derivative from part (a) or substitution and evaluation shown for one x-value on both sides of **one** of *their* stationary points to find the gradients soi or for sketch of any positive cubic.	For 3 marks (or 2 marks) Allow (–2, 52) and $(8, -448)$ to be implied from working if e.g. using 2nd derivative = $6x - 18$, $6(-2) - 18 = -30$ so negative, maximum $6(8) - 18 = 30$ so positive, minimum Condone poor notation for 2nd derivative For **B3** or **B2** accept reasonable sketch drawn in part (c) or referred to if drawn in part (b) accept without axes For **B3** if drawn on axes then max must be in 2nd quadrant and min must be in 4th quadrant For **B2** if drawn on axes allow for max in 2nd quadrant **or** min in 4th quadrant The second derivative must be correct FT from *their* first derivative for **B2** or **M1** May use substitution into ***their*** dy/dx with stated x - values Allow evaluation to 1sig fig rounded or truncated in this context May use substitution into ***their*** dy/dx See AG

Additional Guidance Q5(c)(ii)

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th><th>Partial Marks</th><th>Guidance</th></tr>
<tr><td colspan="5">ALTERNATIVE method using FD’s and equal groups</td></tr>
<tr><td></td><td>Midpoints: 130, 150, 170, 190, 210, 230, 250, 270, 290</td><td>B2</td><td>B1 for 8 correct midpoints</td><td>Accept subdivision into 9 equal intervals of width 20 followed by use of
$Mean = \frac{\Sigma(\text{FD}\times\text{midpoint})}{\Sigma\text{FD}}$</td></tr>
<tr><td></td><td><table><tr><th>Interval</th><th>FD</th><th>m</th><th>FD × m</th></tr><tr><td>120-140</td><td>0.2</td><td>130</td><td>26</td></tr><tr><td>140-160</td><td>0.2</td><td>150</td><td>30</td></tr><tr><td>160-180</td><td>1.1</td><td>170</td><td>187</td></tr><tr><td>180-200</td><td>1.8</td><td>190</td><td>342</td></tr><tr><td>200-220</td><td>1.6</td><td>210</td><td>336</td></tr><tr><td>220-240</td><td>1.6</td><td>230</td><td>368</td></tr><tr><td>240-260</td><td>0.5</td><td>250</td><td>125</td></tr><tr><td>260-280</td><td>0.5</td><td>270</td><td>135</td></tr><tr><td>280-300</td><td>0.5</td><td>290</td><td>145</td></tr><tr><td></td><td>8</td><td></td><td>1694</td></tr></table>
1694 ÷ 8</td><td>M3</td><td>M1 for ΣFDm , m in interval or boundary of interval

M1 dep on first M1 for $\Sigma FDm \div \Sigma FD$</td><td>Condone one further error</td></tr>
<tr><td></td><td>211.75</td><td>B1</td><td></td><td>Condone $200 < v \leq 240$ after 211.75 seen
Accept answers 212, 211.7 or 211.8 nfww</td></tr>
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